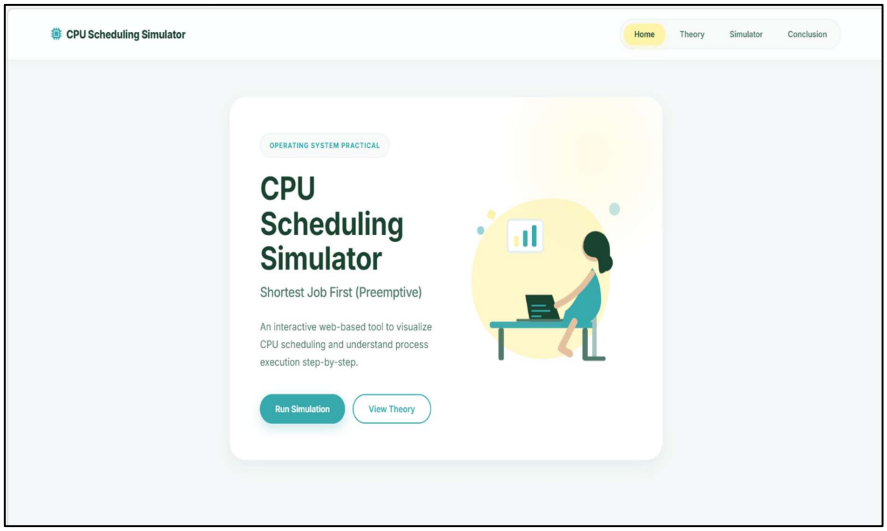
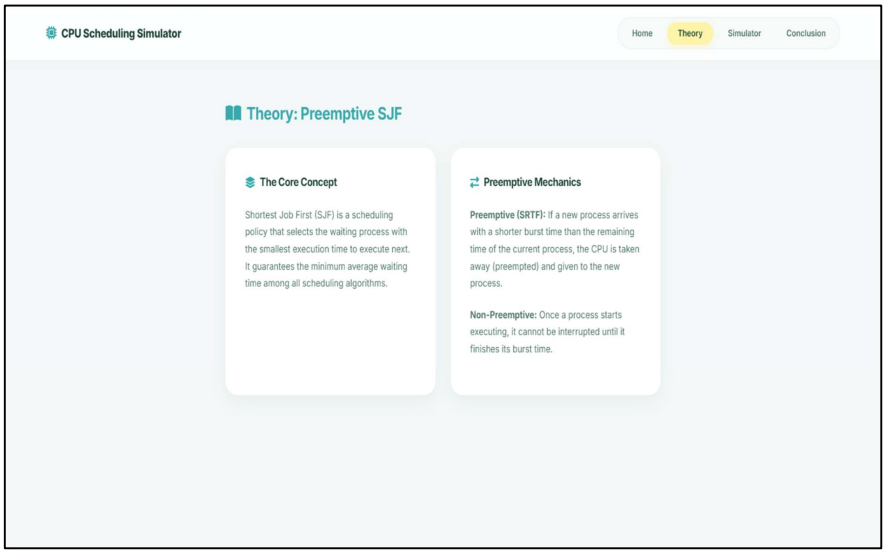
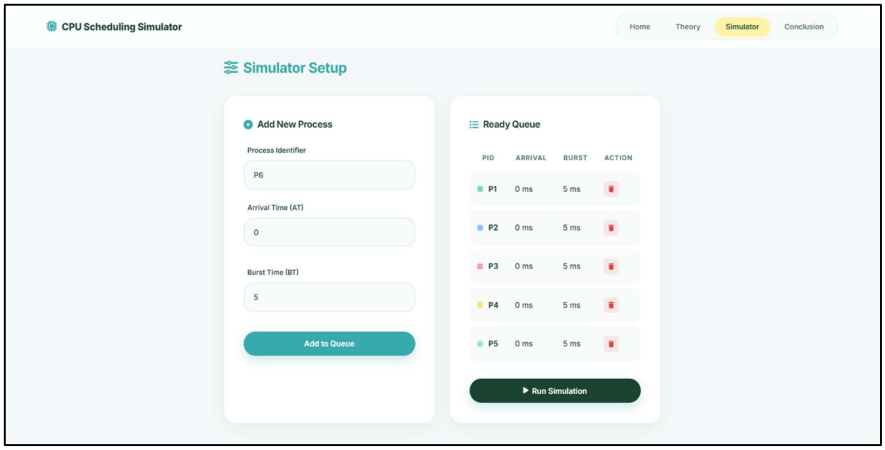
Post lab (open ended):-

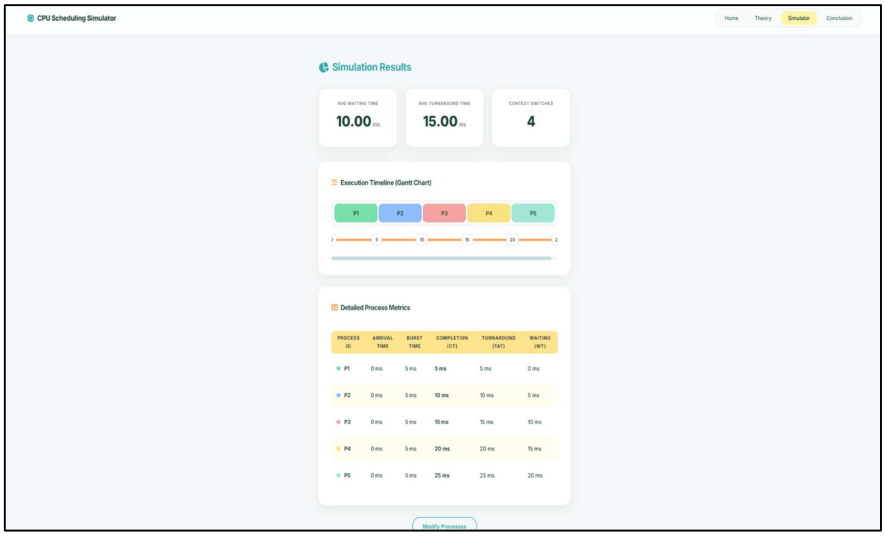
Output:-

Aim: To design and develop an interactive web-based simulator for the Shortest Job First (Preemptive) CPU scheduling algorithm, in order to visualize process execution, calculate scheduling parameters such as waiting time and turnaround time, and understand the working of CPU scheduling in operating systems






CPU Scheduling Simulator
Home
Theory
Simulator
Conclusion
Simulator Setup
Add New Process
Process Identifier
P6
Arrival Time (AT)
0
Burst Time (BT)
5
Add to Queue
Ready Queue
PID ARRIVAL BURST ACTION
P1 0 ms 5 ms
P2 0 ms 5 ms
P3 0 ms 5 ms
P4 0 ms 5 ms
P5 0 ms 5 ms
Run Simulation


CPU Scheduling Simulator
Home
Theory
Simulator
Conclusion
Simulation Results
AVG WAITING TIME
10.00 ms
AVG TURNAROUND TIME
15.00 ms
CONTEXT SWITCHES
4
Execution Timeline (Gantt Chart)
P1
P2
P3
P4
P5
0
5
10
15
20
Detailed Process Metrics
PROCESS ID
ARRIVAL TIME
BURST TIME
COMPLETION (CT)
TURNAROUND (TAT)
WAITING (WT)
P1 0 ms 5 ms 5 ms 5 ms 0 ms
P2 0 ms 5 ms 10 ms 10 ms 5 ms
P3 0 ms 5 ms 15 ms 15 ms 10 ms
P4 0 ms 5 ms 20 ms 20 ms 15 ms
P5 0 ms 5 ms 25 ms 25 ms 20 ms
Modify Processes

CPU Scheduling Simulator
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Analysis & Conclusion
Advantages
Produces the optimal (minimum) average waiting time.
Excellent for systems where throughput is a priority.
Shorter jobs are executed quickly, freeing up memory.
Limitations
Starvation: Long processes may wait indefinitely if short ones keep arriving.
Requires precise prediction of burst times (difficult in reality).
High context switching overhead due to frequent preemptions.